



THE UNIVERSITY OF
MELBOURNE

Machine Learning Applications for Health

COMP90089 (2026) - Lecture 3 - Data Sources

Daniel Capurro





Today's Agenda: Data Sources

- Sources of clinical data
- Some openly (and not so openly) available datasets
- MIMIC



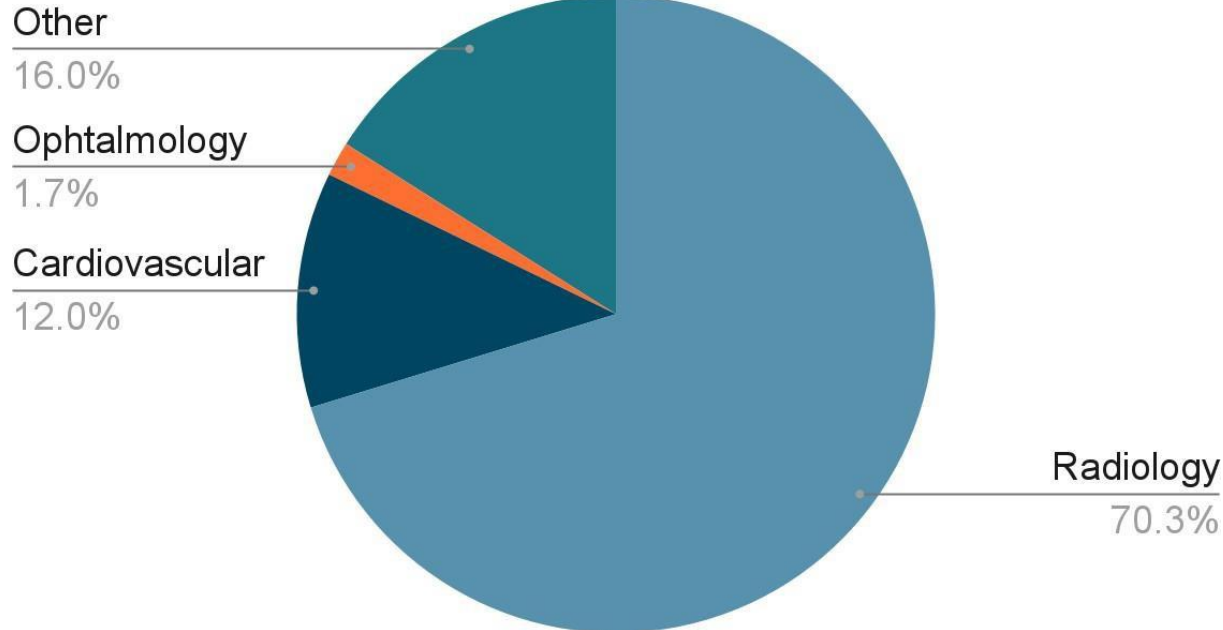
Brainstorming: sources of ~~clinical~~ health data

Which areas of Medicine have made the greatest progress in developing/adopting AI?



Which areas of Medicine have made the greatest progress in adopting AI?

FDA Approvals



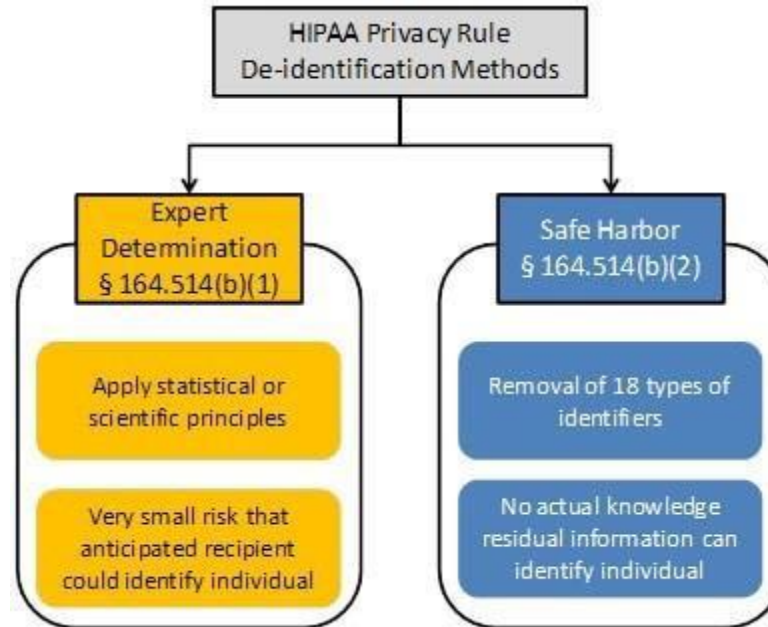


Publicly available data sources

- Not (many) in Australia
- Most publicly datasets are from the USA - HIPAA



In the USA: Health Information Portability and Accountability Act (HIPAA)





Some datasets available

- MIMIC and its companion datasets in Physionet: MANY!!
<https://physionet.org/content/>
- US Medicare SynPUF - <https://www.cms.gov/Research-Statistics-Data-and-Systems/Downloadable-Public-Use-Files/SynPUFs>
- Synthetic data generator - <https://synthetichealth.github.io/synthea/>
- Google public data explorer: <https://datacommons.org/>
- Previous Kaggle challenges: <https://www.kaggle.com/datasets?search=health>
- MTSamples (clinical notes): <https://mtsamples.com/>
- I2b2/n2c2 clinical NLP data sets: <https://n2c2.dbmi.hms.harvard.edu/>

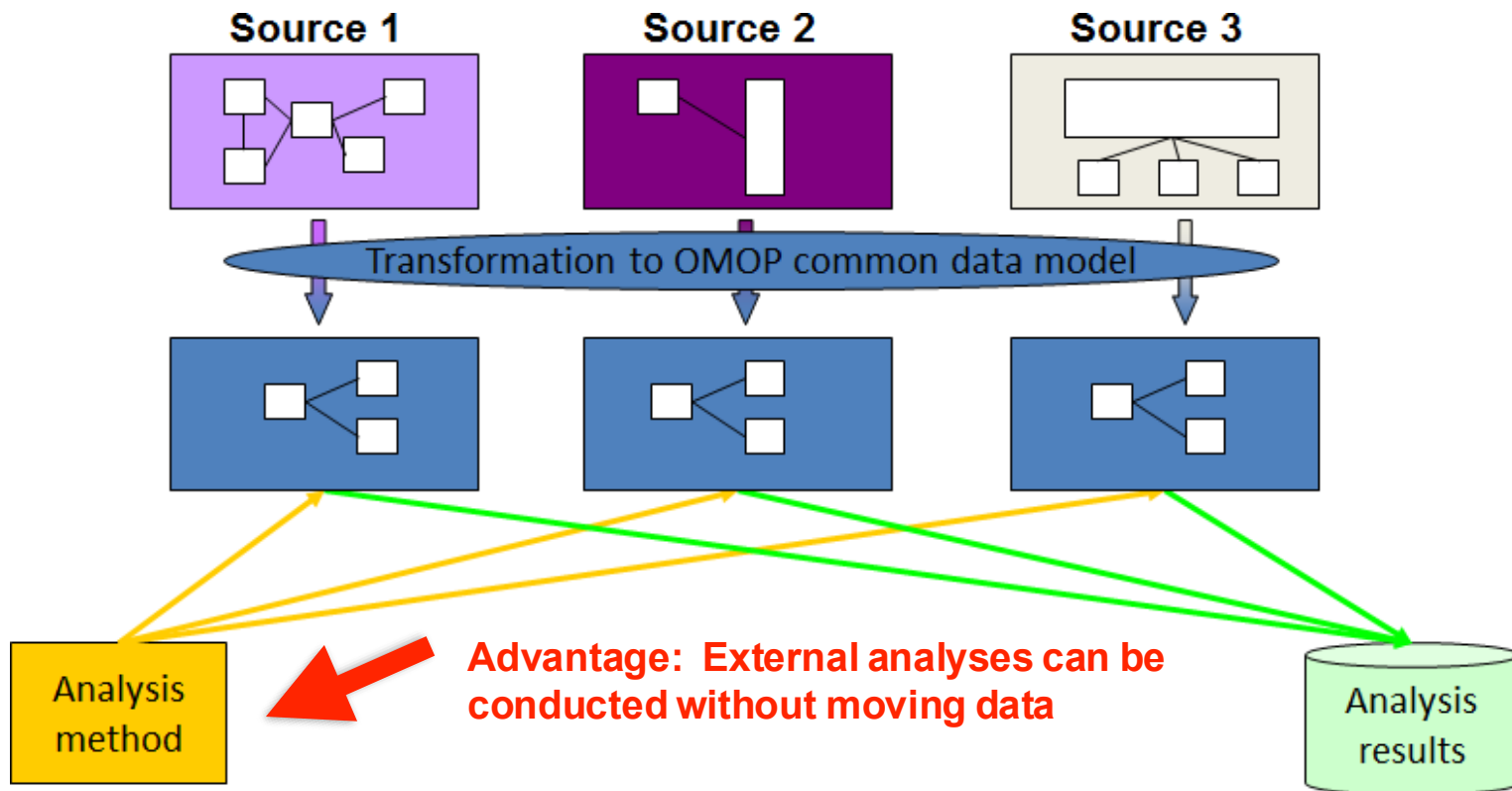


Relevant Australian and Victorian Datasets

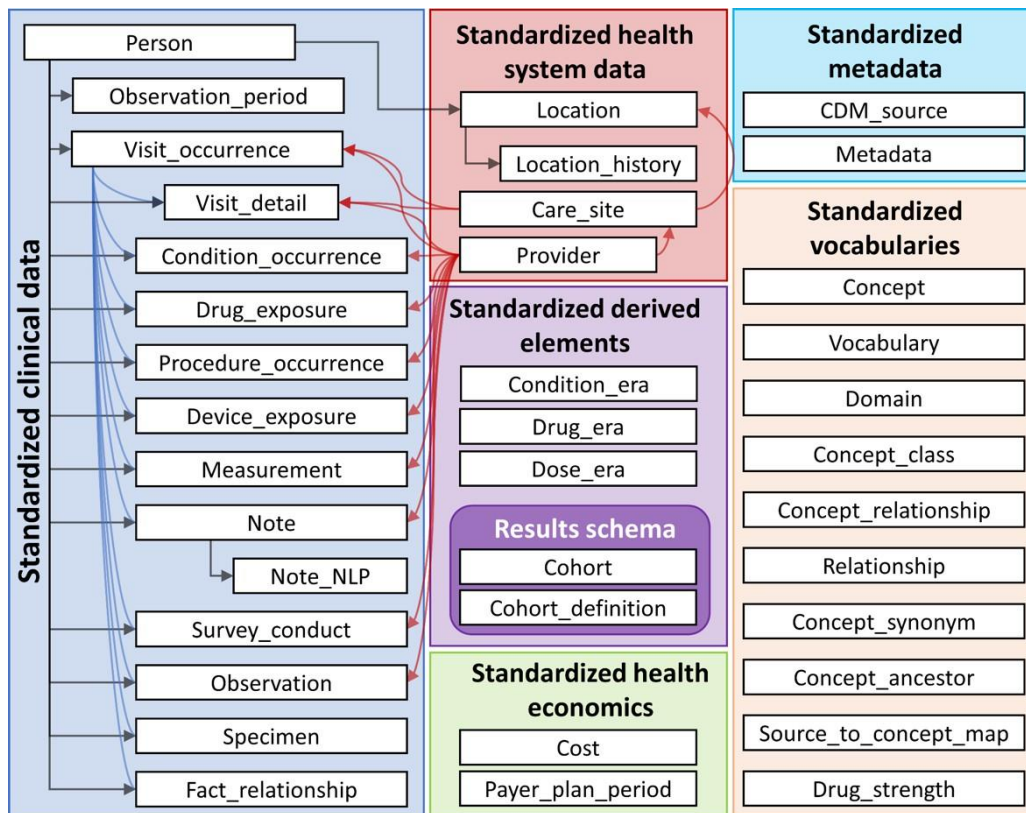
- [Victorian Admitted Episodes Dataset \(VAED\)](#)
- [Victorian Emergency Minimum Dataset \(VEMD\)](#)
- [Victorian Integrated Non-Admitted Health \(VINAH\)](#)
- [Centre for Victorian Data Linkage](#)



A word on Common Data Models



OMOP CDM





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 **N3C** ABOUT ▾ N3C DATA ENCLAVE ▾ GET INVOLVED ▾ RESOURCES ▾ SUPPORT ▾



National COVID Cohort Collaborative

The N3C Data Enclave is a secure platform through which the harmonized clinical data provided by our contributing members is stored. The data itself can only be accessed through a secure cloud portal hosted by NCATS and cannot be downloaded or removed. N3C invites you to begin your journey with the Enclave and join the collaborative efforts of our partners to better understand and address the most pressing COVID-19 clinical questions.

Access the Enclave

Help make science go faster and save lives.

17.3B	1,616.3M	14.8M	5,815,680
Total Rows	Clinical Observations	Persons	COVID+ Cases

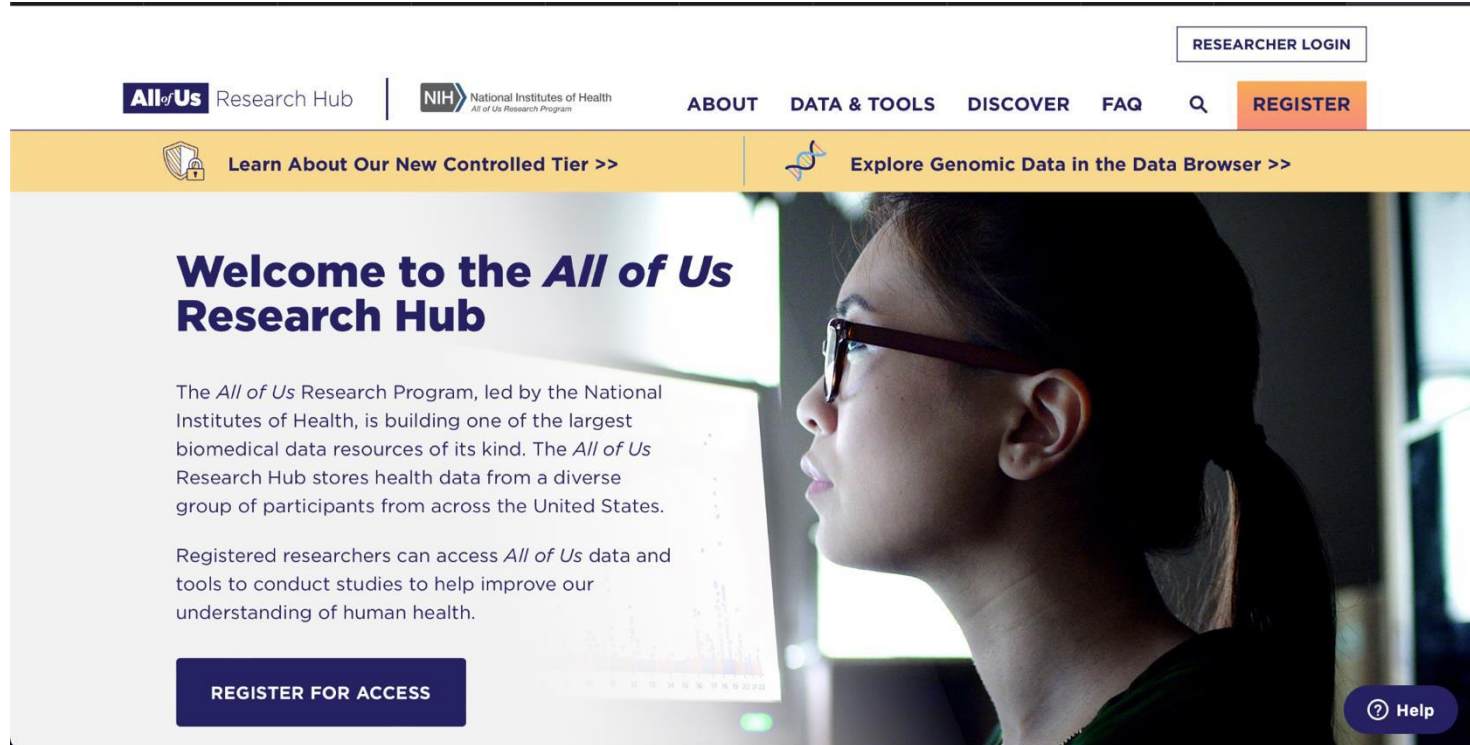
[Explore the Full Cohort Dashboard](#)

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Ask N3C



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A screenshot of the All of Us Research Hub website. The header includes the "All of Us Research Hub" logo, the NIH logo, and navigation links: ABOUT, DATA & TOOLS, DISCOVER, FAQ, a search icon, and a REGISTER button. A RESEARCHER LOGIN button is in the top right. Below the header, there are two yellow banners: "Learn About Our New Controlled Tier >>" with a shield icon and "Explore Genomic Data in the Data Browser >>" with a DNA helix icon. The main content area features a large image of a woman looking at a computer screen. Overlaid on the left is a white box with the heading "Welcome to the All of Us Research Hub", a paragraph about the program, and a "REGISTER FOR ACCESS" button. A "Help" button is in the bottom right corner.

RESEARCHER LOGIN

All of Us Research Hub | NIH National Institutes of Health All of Us Research Program

ABOUT DATA & TOOLS DISCOVER FAQ Q REGISTER

Learn About Our New Controlled Tier >> Explore Genomic Data in the Data Browser >>

Welcome to the *All of Us* Research Hub

The *All of Us* Research Program, led by the National Institutes of Health, is building one of the largest biomedical data resources of its kind. The *All of Us* Research Hub stores health data from a diverse group of participants from across the United States.

Registered researchers can access *All of Us* data and tools to conduct studies to help improve our understanding of human health.

REGISTER FOR ACCESS

Help



Project using OMOP



Welcome to OHDSI Australia

The Observational Health Data Sciences and Informatics (OHDSI, pronounced "Odyssey") initiative is an interdisciplinary collaboration involving various stakeholders. Our mission is to unlock the potential of health data by utilizing large-scale analytics. Rest assured, all our solutions are open-source.

Background

The establishment of OHDSI Australia has been facilitated by close cooperation with the Transformational Data Collaboration (TDC) <https://machaustralia.org/projects/transformational-data-collaboration/>

The TDC operates under the umbrella of the Australian Health Research Alliance (AHRA). Within the AHRA, it falls under the "Data Integration" focus of the "Data Driven Healthcare" activity stream. The TDC is driven by a clear objective:

"To harness the unique, open, and collaborative ethos of AHRA in fostering and supporting national data initiatives where an environment promoting openness, inclusivity, and non-competitiveness is paramount."

Latest News:

[Learn about the EMR to OMOP Project](#)

OHDSI Asia Pacific Symposium Highlights

Recently, the University of New South Wales had the privilege of hosting the OHDSI Asia Pacific Symposium. We were thrilled to see a diverse turnout of clinicians and data scientists from across the globe. For those who couldn't attend or wish to revisit the sessions, recordings are now available.



OMOP Challenges

- Oja et al. transformed Estonian health data (EHR, prescriptions, claims) to OMOP (10% of the population; 150,000 people). 90% of entries successfully mapped (conditions, observations, drugs, measurements)

But there were some challenges!

- Which vocabulary to use? SNOMED-CT? CPT4 (Current Procedural Terminology)?
- Mismatch in granularity (e.g. a text field may say “HPV test”. There are lots of HPV tests (LOINC))
- Time of prescription or time of purchase of a drug? Which to transform to OMOP?

Research and Applications

Transforming Estonian health data to the Observational Medical Outcomes Partnership (OMOP) Common Data Model: lessons learned

Marek Oja , PhD^{1,*}, Sirli Tamm, MSc¹, Kerli Mooses, PhD¹, Maarja Pajusalu, MSc¹, Harry-Anton Talvik, MSc^{1,2}, Anne Ott, MSc¹, Marianna Laht, MD¹, Maria Malk, MSc¹, Marcus Loo, MSc¹, Johannes Holm, MSc¹, Markus Haug, MSc¹, Hendrik Suvalov, MSc¹, Dage Sarg, MSc^{1,2}, Jaak Vilo , PhD^{1,2}, Sven Laur, PhD¹, Raivo Kolde, PhD¹, Sulev Reisberg , PhD^{1,2}

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Ting Dang

ting.dang@unimelb.edu.au

